National Institute of Technology, Rourkela

Name of the Examination: B. Tech. Mid-Semester (Spring 2021-22)

Branch	**: CS**	**Semester**	**: VI**
Title of the Course	**: Computer Networks**	**Course Code**	**: CS3002**
Time: 2 Hours		**Maximum Marks: 30**	

Note : (i) Answer all questions.

(ii) Write all answers of a question at one place.

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1. Answer the following questions:

(a) A large population of ALOHA users manage to generate 50 requests/sec, including both originals and retransmissions. Time is slotted in units of 40 msec. **[3 Marks]**

- (i) What is the chance of success on the first attempt?
- (ii) What is the probability of exactly k collisions and then a success?
- (iii) What is the expected number of transmission attempts needed?

(b) Consider the delay of pure ALOHA versus slotted ALOHA at low load. Which one is less? Explain your answer. **[2 Marks]**

2. Answer the following questions:

(a) Discuss the issues in the data link layer. **[2 Marks]**

(b) Explain the key differences between OSI and TCP/IP reference models? **[3 Marks]**

3. Answer the following questions:

(a) Explain the channel allocation methods for broadcast networks. Give advantages and disadvantages of these methods? **[3 Marks]**

(b) A disadvantage of a broadcast subnet is the capacity wasted when multiple hosts attempt to access the channel at the same time. As a simplistic example, suppose that time is divided into discrete slots, with each of *n* hosts attempting to use the channel with probability *p* during each slot. What fraction of the slots will be wasted due to collisions? **[2 Marks]**

4. Answer the following questions:

(a) Sixteen-bit message are transmitted using a Hamming code. How many check bits are needed to ensure that the receiver can detect and correct single-bit errors? Show the bit pattern transmitted for the message 1101001100101101. Assume that even-parity is used in the Hamming code. **[3 Marks]**

(b) What is the maximum overhead in byte-stuffing and bit-stuffing algorithms? **[2 Marks]**

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5. Answer the following questions:

(a) Explain the following two protocols: **[3 Marks]**

(i) A Bit-Map Protocol

(ii) The Binary Countdown

(b) Give the differences between the CSMA and Adaptive Tree Walk Protocol? **[2 Marks]**

6. Answer the following questions: **[1.5 + 1.5 + 2 = 5 Marks]**

(a) Explain the differences between the Simplex protocols and Sliding window protocols?

(b) Explain the drawback of 1-bit sliding window protocol?

(c) Explain the key differences between Token Bus and Token Ring networks?